$$= 1 + 2\sin\left(\frac{\pi}{4} - \frac{C}{4}\right)\left[\cos\left(\frac{B-A}{4}\right) - \sin\left(\frac{A+B}{4}\right)\right]$$

$$= 1 + 2\sin\left(\frac{\pi}{4} - \frac{C}{4}\right)\left[\cos\left(\frac{B-A}{4}\right) - \cos\left(\frac{\pi}{2} - \frac{A+B}{4}\right)\right]$$

$$= 1 + 2\sin\left(\frac{\pi}{4} - \frac{C}{4}\right)\left[2\sin\left(\frac{\frac{B-A}{4} + \frac{\pi}{2} - \frac{A+B}{4}}{2}\right) - \sin\left(\frac{\frac{\pi}{2} - \frac{A+B}{4} - \frac{B-A}{4}}{2}\right)\right]$$

$$= 1 + 4\sin\left(\frac{\pi - A}{4}\right)\sin\left(\frac{\pi - B}{4}\right)\sin\left(\frac{\pi - C}{4}\right)$$

எடுத்துக்காட்டு 3.41 $A + B + C = \pi$ **எனில்,** $\cos^2 A + \cos^2 B + \cos^2 C = 1 - 2\cos A \cos B \cos C$ **என நிறுவுக.**

தீர்வு:

$$\cos^2 A + \cos^2 B + \cos^2 C = \frac{1}{2}[2\cos^2 A + 2\cos^2 B + 2\cos^2 C]$$

$$= \frac{1}{2}[(1 + \cos 2A) + (1 + \cos 2B) + (1 + \cos 2C)]$$

$$= \frac{3}{2} + \frac{1}{2}(\cos 2A + \cos 2B + \cos 2C)$$

$$= \frac{3}{2} + \frac{1}{2}[2\cos(A + B)\cos(A - B) + (2\cos^2 C - 1)]$$

$$= \frac{3}{2} + \frac{1}{2}[-2\cos C\cos(A - B) + 2\cos^2 C - 1] \quad (\because A + B = \pi - C)$$

$$= \frac{3}{2} - \frac{1}{2} + \frac{1}{2}[-2\cos C(\cos(A - B) - \cos C)]$$

$$= 1 - \cos C[\cos(A - B) - \cos C]$$

$$= 1 - \cos C[\cos(A - B) + \cos(A + B)]$$

$$= 1 - \cos C[2\cos A\cos B]$$

$$= 1 - 2\cos A\cos B\cos C$$

பயிற்சி 3.7

1. $A + B + C = 180^\circ$ **எனில், பின்வருவனவற்றை நிறுவுக.**

(i) $\sin 2A + \sin 2B + \sin 2C = 4\sin A\sin B\sin C$

(ii) $\cos A + \cos B - \cos C = -1 + 4\cos\frac{A}{2}\cos\frac{B}{2}\sin\frac{C}{2}$

(iii) $\sin^2 A + \sin^2 B + \sin^2 C = 2 + 2\cos A\cos B\cos C$

(iv) $\sin^2 A + \sin^2 B - \sin^2 C = 2\sin A\sin B\cos C$

(v) $\tan\frac{A}{2}\tan\frac{B}{2} + \tan\frac{B}{2}\tan\frac{C}{2} + \tan\frac{C}{2}\tan\frac{A}{2} = 1$

(vi) $\sin A + \sin B + \sin C = 4\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$

(vii) $\sin(B + C - A) + \sin(C + A - B) + \sin(A + B - C) = 4\sin A\sin B\sin C$

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